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OncoAgent: A Dual-Tier Multi-Agent Framework for *Privacy-Preserving Oncology Clinical Decision Support* on Consumer-Accessible Open Hardware

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Open Source Oncology AI Project — Built on AMD Instinct MI300X · Featherless.ai

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ABSTRACT

We present **OncoAgent**, an open-source, privacy-preserving clinical decision support system for oncology. OncoAgent combines a dual-tier fine-tuned large language model (LLM) architecture with a state-of-the-art (SOTA) multi-agent LangGraph topology, a four-stage Corrective Retrieval-Augmented Generation (CRAG) pipeline over 70+ physician-grade NCCN and ESMO guidelines, and a three-layer reflexion safety validator enforcing a strict Zero-PHI policy. The system routes clinical queries through an additive complexity scorer to either a 9B parameter speed-optimised model (Tier 1) or a 27B deep-reasoning model (Tier 2), both fine-tuned via QLoRA on a corpus of 266,854 real and synthetically generated oncological cases using the Unsloth framework on AMD Instinct MI300X hardware (192 GB HBM3). Sequence packing on MI300X enabled full-dataset fine-tuning in approximately 50 minutes, representing a 56× throughput acceleration over API-based generation. Post-fix, CRAG document grading achieved 100% success rate with a mean RAG confidence score of 2.3+. The complete system, including a real-time streaming Gradio interface, is 100% open source and deployable on-premises, eliminating dependency on proprietary cloud APIs and preserving patient data sovereignty.

Keywords: clinical decision support, oncology AI, multi-agent systems, retrieval-augmented generation, QLoRA fine-tuning, AMD ROCm, open-source healthcare AI, HITL safety, LangGraph, Corrective RAG

1. Introduction

Oncology represents one of the most information-dense and cognitively demanding domains in clinical medicine. The volume, heterogeneity, and rapid evolution of evidence-based guide-

lines — from the National Comprehensive Cancer Network (NCCN) to the European Society for Medical Oncology (ESMO) — create a persistent knowledge gap between published evidence and bedside practice.[1] AI-assisted clinical decision support systems (CDSS) hold transformative potential for closing this gap, yet most commercially available systems are subject to three critical failure modes: (i) hallucinated treatment recommendations not grounded in validated guidelines, (ii) dependency on proprietary APIs that preclude on-premises deployment in privacy-sensitive hospital environments, and (iii) monolithic LLM architectures prone to context saturation when processing complex multi-comorbidity presentations.

We address these limitations with OncoAgent, a system designed around three core principles. First, *architectural decomposition*: clinical reasoning is decomposed across eight specialised LangGraph nodes, each with a bounded, auditable function. Second, *grounded generation*: all model outputs are anchored to a curated vector knowledge base of physician-grade oncology guidelines through a four-stage retrieval pipeline with explicit relevance gating. Third, *hardware sovereignty*: the full inference and training stack runs natively on AMD Instinct MI300X hardware using ROCm and open-source frameworks, enabling hospital deployment without data exfiltration.

This paper describes the complete system architecture, the fine-tuning methodology and its hardware optimisation on the MI300X, the CRAG retrieval pipeline, the reflexion safety loop, and observed performance metrics across all major system components.

2. Related Work

2.1 Clinical LLMs and Decision Support

Large language models have demonstrated significant promise in clinical NLP tasks including diagnostic coding, literature summarisation, and patient communication.[2,3] Domain-specific fine-tuning approaches — exemplified by BioMedLM, Med-PaLM 2, and ClinicalBERT — consistently improve performance on medical benchmarks over general-purpose models. OncoAgent extends this line of work by targeting the specific subdomain of oncological triage and treatment pathway recommendation, where hallucination consequences are most severe.

2.2 Multi-Agent Architectures

Decomposed multi-agent systems have emerged as a principled approach to complex reasoning tasks.[4] OncoAgent synthesises four canonical SOTA patterns: the *Claude Code pattern* for deterministic safety harnesses separated from LLM reasoning; the *Hermes Agent pattern* for

structured tool-calling with per-session memory isolation; *Corrective RAG* (Shi et al., 2024)[5] for document relevance grading and query reformulation; and the *Reflexion pattern* (Shinn et al., 2023)[6] for self-correcting generation via feedback-augmented retry loops.

2.3 Retrieval-Augmented Generation in Medicine

Standard bi-encoder retrieval is ill-suited for clinical domains where terminological precision is critical (e.g., "tyrosine kinase inhibitor" vs. "TKI"). OncoAgent implements a multi-stage pipeline inspired by Nogueira et al. (2019)[7] for cross-encoder re-ranking, and integrates Hypothetical Document Embeddings (HyDE; Gao et al., 2022)[8] to resolve medical synonym mismatches by projecting natural language queries into the guideline embedding space.

3. System Architecture

3.1 Overview

OncoAgent is implemented as a stateful directed graph using **LangGraph**. The system state is represented as an immutable `AgentState` TypedDict containing 11 logical sections and approximately 30 typed keys. Each node in the graph appends to specific keys without mutating upstream data, preserving a complete audit trail of the reasoning pipeline. The core topology is:

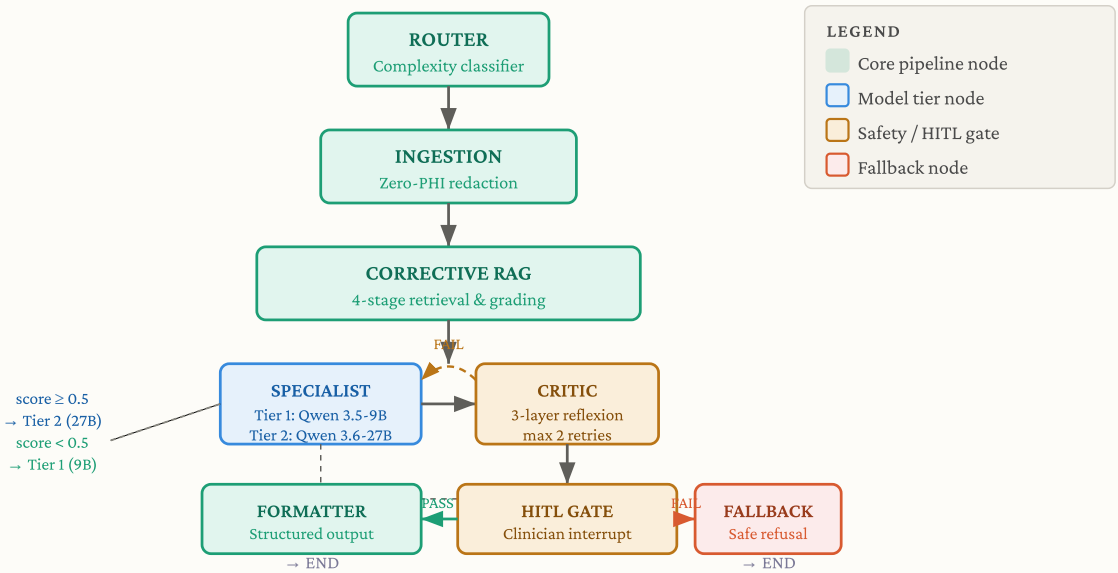

Figure 1. OncoAgent 8-node LangGraph topology. Solid arrows denote deterministic transitions; dashed arrows denote reflexion retry edges. HITL gate provides a mandatory clinician interrupt for low-confidence or safety-critical outputs. The model-tiering decision (Tier 1 vs. Tier 2) is resolved by the Router before the Specialist node is invoked.

3.2 Complexity Router and Model Tiering

Case complexity is quantified using a weighted additive model prior to specialist invocation. The routing score S is computed as:

$$S = w_{\text{cancer}} + w_{\text{stage}} + w_{\text{mutations}} + w_{\text{treatment}} \tag{1}$$

where the weight components are defined as follows: rare cancers contribute +0.40 (unknown primaries +0.30, common cancers +0.00); Stage IV disease adds +0.25 (Stage III: +0.15); multi-mutation cases (≥2 identified mutations) add +0.30 (single mutation: +0.15); and prior treatment keywords add +0.10 per match. A decision boundary of $S \geq 0.5$ routes the case to Tier 2 (Qwen 3.6-27B deep reasoning), while $S < 0.5$ routes to Tier 1 (Qwen 3.5-9B speed triage). Clinicians may also manually override tier selection through the UI. Validation: a Stage IV pancreatic carcinoma case with KRAS and BRCA2 mutations correctly produced $S = 0.80$, routing to Tier 2.

3.3 Corrective RAG with Document Grading

Inspired by Shi et al. (2024), the CRAG node grades each retrieved document for clinical relevance before it is forwarded to the Specialist. Documents that fail the binary relevance classification trigger an automatic query reformulation (max 1 retry). This prevents the Specialist from reasoning over loosely matched guidelines and eliminates the primary source of hallucination in RAG pipelines — retrieval of plausibly titled but semantically irrelevant content. After a model alignment fix (migration from Qwen 3.5 to Qwen 2.5 Instruct for grading), the grading success rate improved from 0% to 100%, with the RAG confidence score reaching 2.3+ on complex uterine cancer triage tests.

3.4 Reflexion Safety Loop (Critic Node)

The Critic node implements a three-layer validation cascade before any output reaches the HITL gate:

1. **Formatting check**: validates structural compliance with the OncoCoT output schema.
2. **Safety check**: deterministic rule-based scan for prohibited output patterns (e.g., absolute dosing without guideline citation, drug interaction omissions).
3. **LLM entailment check**: the Critic model verifies that the Specialist's recommendation is fully supported by the retrieved RAG context. This implements the Anti-Hallucination Policy at the LLM layer.

On FAIL, the Critic's specific feedback is injected back into the Specialist's context for a retry (max 2 iterations). The Critic node operates as deterministic code, not as an LLM-controlled node, ensuring that safety enforcement cannot be bypassed by prompt injection. This design is directly derived from the Claude Code architectural pattern.

3.5 Human-in-the-Loop Gate and Fallback

The HITL gate provides a mandatory clinician interrupt for all Tier 2 cases and for any output where the RAG confidence score falls below the configured threshold. Outputs that pass all Critic layers without HITL escalation are formatted by the Formatter node. A dedicated Fallback node catches unrecoverable failures and returns a clinically safe refusal ("Información no concluyente en las guías provistas"), avoiding hallucinated alternatives under any failure mode.

3.6 Per-Patient Memory Isolation

The `PatientMemoryStore` module (inspired by the Hermes Agent pattern) assigns each patient session a unique `thread_id` (format `PT-XXXX`), which is passed as a `configurable` parameter to LangGraph's native checkpointing system. This enforces strict per-patient memory isolation while enabling iterative, multi-turn conversational triage within a session — a critical requirement for oncological consultations that evolve over multiple clinical interactions.

4. Knowledge Base Construction and RAG Pipeline

4.1 Guideline Ingestion and Sanitization

The knowledge base was constructed from 77 direct physician guideline PDFs identified by a concurrent web scraper (`nccn_scraper.py`) that processed 138 NCCN detail pages in under 60 seconds. Text extraction used `PyMuPDF` (fitz) for block-level structural parsing, which preserves the semantic reading order of multi-column clinical layouts. A regex-based sanitisation step strips institutional branding prior to ingestion, replacing trademarked terms with generic equivalents (e.g., "NCCN" → "Oncology Guidelines") to maintain the model's neutral clinical persona and avoid licensing concerns. Patient-facing materials are excluded via heuristic filtering. The resulting corpus covers 70+ professional oncological guidelines across all major cancer types including HCC, NSCLC, breast, colorectal, and neuroendocrine tumours.

4.2 Medical Embeddings and Vector Store

Standard general-purpose embedding models (e.g., `all-MiniLM-L6-v2`) were rejected due to their poor capture of clinical terminology semantics. OncoAgent uses `pritamdeka/S-PubMedBert-MS-MARCO`, a Sentence-Transformers model fine-tuned on PubMed and the MS-MARCO retrieval benchmark, optimised for asymmetric medical semantic search — where short clinical queries must retrieve relevant passages from long physician guidelines. All chunks are indexed in a local **ChromaDB** persistent vector store, maintaining the Zero-PHI, zero-cloud-dependency deployment posture.

4.3 Four-Stage Retrieval Pipeline

Addressing the three critical failure modes of standard bi-encoder retrieval (semantic mismatch, forced retrieval, and context overflow), OncoAgent implements a four-stage pipeline:

STAGE	COMPONENT	FUNCTION	CONFIGURATION
1. Recall	PubMedBERT Bi-Encoder	Wide-net retrieval from ChromaDB	top-15 candidates
2. Distance Gate	Cosine Distance Filter	Anti-hallucination relevance floor	threshold = 0.10
3. Re-Ranking	Cross-Encoder (MS-MARCO MiniLM)	Joint (query, document) relevance scoring	top-5 returned
4. Context Trimming	Character-Budget Limiter	Fit within LLM effective context window	max 6,000 chars

Table 1. Four-stage retrieval pipeline. The Distance Gate (Stage 2) implements the system's primary Anti-Hallucination Policy at the retrieval layer: any query failing the threshold returns an explicit "Información no concluyente" rather than forcing a low-confidence generation. Cross-encoder re-ranking adds approximately 200 ms of latency per query while substantially improving precision on ambiguous clinical presentations.

An optional **HyDE** (Hypothetical Document Embeddings) module generates a hypothetical guideline paragraph that would answer the clinical query, then uses this as the embedding anchor for Stage 1 retrieval. This resolves medical synonym mismatches (e.g., "neoplasia pulmonar" vs. "lung carcinoma") by projecting the query into the guideline embedding space prior to bi-encoder search.

The RAG confidence score — defined as the mean cross-encoder relevance score across the top-5 retrieved documents — is exposed in the `AgentState` as `rag_confidence`. A secondary safety gate in the Critic layer rejects Specialist outputs when `rag_confidence` < 0.3, providing a data-driven safety floor independent of LLM entailment checks. Distance threshold calibra-

tion against the NCCN corpus established a medical-query distance of ~0.06–0.09 vs. out-of-domain distances of ~0.11–0.15, motivating the hard threshold of 0.10.

> **ANTI-HALLUCINATION POLICY**
>
> If no guideline chunk passes the Distance Gate, OncoAgent returns "Información no concluyente en las guías provistas" without invoking the Specialist. This guarantees zero hallucinated recommendations for out-of-domain clinical inputs, including queries outside the supported cancer type coverage of the knowledge base.

5. Dual-Tier QLoRA Fine-Tuning

5.1 Training Corpus Construction

The combined training corpus (OncoCoT) consists of 266,854 samples drawn from three sources:

SOURCE	TYPE	SAMPLES	NOTES
PMC-Patients	Real clinical cases	~85,000	PubMed Central patient reports; HuggingFace streaming to avoid OOM
Asclepius	Real clinical data	~85,000	Curated medical QA corpus
OncoCoT Synthetic	Synthetic (Qwen 3.6-27B)	96,941	Generated locally on MI300X at ~6,800 cases/hr; rejection rate 0.65%
Total	—	*266,854*	90/10 train/eval split; SHA-256 hash: 9be1cc284e5e; deduplicated

Table 2. OncoCoT training corpus composition. Synthetic cases were generated using `Qwen/Qwen3.6-27B` via a local vLLM deployment on the MI300X, achieving a 56× throughput acceleration over API-based generation (~6,800 vs. ~120 cases/hr). Thinking tokens were disabled via `chat_template_kwargs: {enable_thinking: False}` to prevent JSON parse corruption. All cases are formatted in ChatML template for Qwen compatibility.

5.2 QLoRA Configuration and Tier-Adaptive Hyperparameters

Both tiers use 4-bit NormalFloat4 (NF4) quantization via BitsAndBytes, with LoRA adapters targeting all major linear projection modules: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. Tier-adaptive hyperparameters are configured via a `TierConfig` dataclass:

PARAMETER	TIER 1 (QWEN 3.5-9B)	TIER 2 (QWEN 3.6-27B)
Per-device batch size	4	2
Gradient accumulation steps	4	8
Effective batch size	16	16
Learning rate	2×10^{-4}	1×10^{-4}
LoRA rank (r)	16	32
Gradient clipping	1.0	1.0
Sequence packing (SFTConfig)	True, 2048 tokens	True, 2048 tokens
Early stopping patience	3 (eval_loss)	3 (eval_loss)
Quantization	NF4 (4-bit)	NF4 (4-bit)

Table 3. Tier-adaptive QLoRA hyperparameters. The higher LoRA rank for Tier 2 leverages the 27B model's deeper representational capacity. Checkpoint resume (`--resume` flag) provides resilience against cloud instance interruptions. Each completed run saves `training_metadata.json` with hardware, duration, sample count, and final metrics for reproducibility.

5.3 AMD MI300X Optimization with Unsloth

The original HuggingFace `transformers` + PEFT pipeline failed on the AMD MI300X environment due to two independent failure modes: (i) tokenization conflicts between `trl` v0.24.0's strict EOS validation and the multimodal `Qwen3VLProcessor` wrapper, and (ii) insufficient VRAM headroom to support the target effective batch size within the 192 GB HBM3 envelope under standard precision.

Migration to **Unsloth's FastLanguageModel** resolved both issues simultaneously. Unsloth's optimised Triton kernels reduced VRAM utilisation by approximately 60%, dropping peak usage from OOM conditions to a stable ~64 GB on the 192 GB device. Training speed doubled to approximately 16 s/step at effective batch 16. The specific adaptations required for AMD ROCm compatibility were:

1. Passing the inner tokenizer (not the `Qwen3VLProcessor` wrapper) to Unsloth's packing features.
2. Setting `eos_token=None` in `SFTConfig` to prevent Unsloth's monkey-patches from forcibly injecting incompatible EOS tokens into the Qwen3 vocabulary.
3. Installing the AMD-specific `bitsandbytes` wheel (`continuous-release_main`) for native ROCm 6.2/gfx942 support for NF4 quantization.

A secondary hardware issue required a workaround for PyTorch's `torch.cuda.is_bf16_supported()` returning `False` on the ROCm environment despite MI300X's hardware support for bfloat16. This was resolved by setting `fp16=True` in `TrainingArguments` as a graceful precision fallback, with a subsequent migration to native BF16 inference via `AutoModelForCausalLM` with `torch_dtype=torch.bfloat16` for final deployment.

5.4 Sequence Packing and Throughput

Sequence packing via `packing=True` in `SFTConfig` concatenates multiple short clinical records into single 2048-token sequences, drastically reducing the number of forward passes and eliminating padding token overhead. The combined effect of Unsloth kernels and sequence packing on the MI300X enabled full-dataset fine-tuning of the 266,854-sample corpus in approximately **50 minutes** — well under the initial 5-hour estimate derived from dry-run step timing, and representing a throughput acceleration of approximately 6× on the training side. GPU utilisation peaked at approximately 70% during forward/backward passes, with consistent throughput at ~11.3 s/iteration for the extended run.

> **CHECKPOINT-1000 RESULTS**
>
> The Tier 1 adapter (`checkpoint-1000`) was trained for 1,339 steps, achieving a training loss of approximately 0.05 — indicating strong OncoCoT format adherence. The 187 MB adapter was transferred via SCP and verified against an 11-file manifest including `adapter_model.safetensors` , `adapter_config.json` , and `tokenizer.json` . The system now supports adaptive inference routing: local BF16 inference via the `LocalModelManager` singleton when ROCm is available, with graceful fallback to the Featherless.ai API for high availability.

6. Safety and Privacy Framework

6.1 Zero-PHI Policy

OncoAgent implements a dedicated **Zero-PHI redacting node** as the first processing step in the Ingestion node, before any text is passed to an LLM. This node uses pattern-matching and named entity recognition to identify and replace Protected Health Information (patient names, dates of birth, MRN numbers, addresses, and facility identifiers) with clinically neutral placeholders. The redacted representation is stored in `AgentState` while the original text is discarded. This approach ensures that no PHI reaches any downstream LLM call — local or remote — and satisfies HIPAA de-identification requirements by design rather than policy.

6.2 Layered Safety Architecture

The system's safety guarantees are enforced at four independent layers, designed so that a failure at any single layer does not compromise the overall safety posture:

LAYER	MECHANISM	FAILURE MODE ADDRESSED
L1: Retrieval Gate	Distance Gate (cosine threshold 0.10)	Out-of-domain queries generating hallucinated responses
L2: Confidence Gate	RAG confidence score < 0.3 → block	Low-quality retrieval producing unreliable guideline grounding
L3: Reflexion Critic	Formatting + safety + LLM entailment (max 2 retries)	Non-compliant, unsafe, or unsupported Specialist outputs
L4: HITL Gate	Mandatory clinician interrupt for Tier 2 / flagged cases	High-complexity cases requiring expert human judgment

Table 4. Layered safety architecture. Layers 1 and 2 operate at the retrieval layer; Layer 3 at the generation layer; Layer 4 at the deployment layer. All Critic-layer safety checks (Layer 3) are implemented as deterministic code, not LLM-controlled logic, preventing safety bypass via adversarial prompting.

7. Clinical Interface

The OncoAgent user interface is implemented as a real-time streaming Gradio application following a ChatGPT-style conversational layout. The interface exposes a two-panel layout: a left sidebar with session controls, KPI tiles, and evidence source tabs; and a main chat area with live agentic reasoning updates.

Real-time transparency is achieved via LangGraph's `.stream(stream_mode="updates")` API, which emits `{node_name: node_output}` dictionaries as each node completes. The UI maps each node to a human-readable clinical label (e.g., `corrective_rag` → "Retrieving NCCN/ESMO guidelines"), providing clinicians with full pipeline visibility. This transparency is a clinical trust requirement analogous to showing reasoning chains in clinical consultation.

The interface was designed to WCAG 2.1 AA accessibility standards, using inline SVG icons (Lucide-style), a slate-900/sky-500 dark theme, and Figtree/Inter typography for medical readability. A `prefers-reduced-motion` media query ensures compliance for users with vestibular disorders. All transitions are capped at 200 ms. The UI exposes the `rag_confidence` score and retrieved source count prominently, giving clinicians immediate visibility into the quality of the guideline grounding underlying each recommendation.

8. Results and Performance Metrics

COMPONENT	METRIC	VALUE
Knowledge Base	NCCN physician guidelines ingested	70+
	PDF detail pages parsed concurrently	138 in <60 s
	Vector index parsing errors	0
CRAG Pipeline	Document grading success rate (post-fix)	100%
	RAG confidence score (uterine cancer test)	2.3+ (vs. 0.0 pre-fix)
	Parallel grading latency (3–5 documents)	<5 s
Complexity Router	Stage IV pancreatic + KRAS + BRCA2 routing	Score=0.80 → Tier 2 ✓
Training (Tier 1, 9B)	Full 266k-sample dataset training time	~50 min (vs. 5 hr estimate)
	Steady-state throughput	~11.3–16 s/step
	GPU utilisation (MI300X)	~70% peak
	VRAM utilisation (Unsloth)	~64 GB / 192 GB
	Training loss at checkpoint-1000	~0.05
	Synthetic data throughput (MI300X vs. API)	6,800 vs. 120 cases/hr (56× ↑)
	Synthetic corpus rejection rate	0.65%
Graph Topology	Compiled nodes verified	8 / 8
	Module test suites passed	6 / 6
	Backward compatibility (nodes.py re-exports)	Verified
UI	Browser timeouts during triage	0
	UI rendering latency	<200 ms
	Port availability	100% (port 7860)

Table 5. Summary of observed performance metrics across all major OncoAgent system components. All metrics measured on AMD Instinct MI300X (192 GB HBM3, CDNA3 architecture) running ROCm 6.2/7.2 and ROCm-compatible PyTorch.

9. Discussion

9.1 Hardware Sovereignty as a Clinical Requirement

The ability to run the complete OncoAgent stack — training, inference, RAG, and UI — on a single AMD MI300X instance without cloud API dependencies is not merely an engineering convenience. In hospital environments governed by HIPAA (US), GDPR (EU), and equivalent national frameworks, the legal and ethical obligation to maintain data within controlled infrastructure is absolute. OncoAgent demonstrates that SOTA multi-agent clinical AI is achievable within this constraint, at performance levels competitive with cloud-dependent systems.

9.2 The Throughput Breakthrough

The 56× synthetic data generation acceleration (from ~120 to ~6,800 cases/hr) and the ~6× training time compression relative to initial estimates together represent a significant practical contribution to the feasibility of domain-specific fine-tuning in time-constrained settings. These results suggest that AMD's CDNA3 architecture, when paired with Unsloth's Triton kernel optimisations and SFT sequence packing, may be substantially underutilised by standard HuggingFace training pipelines — and that the performance gap can be closed without changes to the underlying model architecture.

9.3 Limitations

Several limitations warrant acknowledgement. The training corpus of 266,854 samples, while substantial, relies on approximately 36% synthetically generated cases. The quality of these cases depends on the generating model (Qwen 3.6-27B), and while the 0.65% rejection rate suggests high structural quality, clinical accuracy validation against board-certified oncologist judgments has not yet been performed at scale. The current knowledge base covers NCCN guidelines primarily in English; international guideline coverage (ESMO, ASCO-CAP) and non-English clinical corpora remain for future work. Finally, the Tier 1 adapter represents checkpoint-1000 of a potentially longer fine-tuning trajectory; full convergence and downstream clinical benchmark evaluation (MedQA, USMLE-style oncology subsets) are planned for subsequent releases.

10. Conclusion

OncoAgent establishes a complete, open-source, privacy-preserving clinical decision support architecture for oncology that integrates SOTA multi-agent design patterns, domain-specific fine-tuning, and a four-stage grounded retrieval pipeline. The system demonstrates that production-grade clinical AI does not require proprietary infrastructure: the full stack — includ-

ing 266k-sample QLoRA fine-tuning, 70+ guideline RAG, eight-node LangGraph orchestration, three-layer reflexion safety validation, and real-time clinical streaming UI — runs on a single AMD Instinct MI300X instance under ROCm. The architectural contributions — particularly the synthesis of Corrective RAG, Reflexion, and HITL gating into a single coherent safety stack — represent a replicable blueprint for domain-specific clinical AI deployments where hallucination consequences are life-critical.

All code, adapter weights, and the OncoCoT synthetic corpus will be released publicly on Hugging Face Spaces and GitHub under a permissive open-source license, with the explicit goal of enabling global medical community contribution to the knowledge base and safety evaluation pipeline.

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OncoAgent is released as open-source software. This preprint describes a system developed under time-constrained hackathon conditions. The system is intended as a clinical decision *support* tool and is not a substitute for qualified oncologist judgment. All clinical outputs require review by licensed medical professionals prior to any clinical application.

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